

UNIVERSITY OF RHODE ISLAND

Data Science at URI

Interdisciplinary, hands-on, career-ready

Open house

What is Data Science?

Turn data into useful information

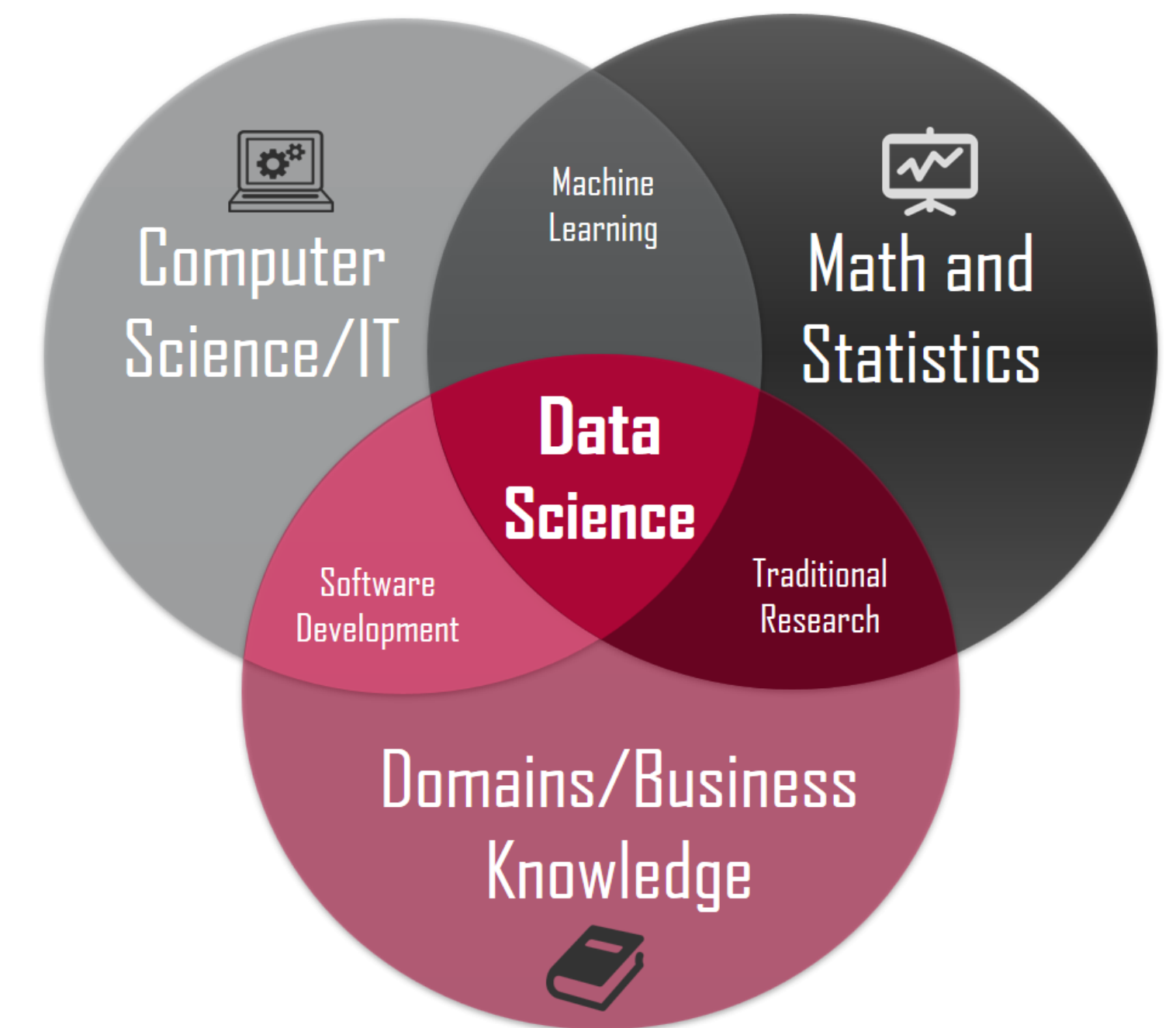
- DS is the study of *data* to *extract meaningful insights*.

Multidisciplinary study

- DS is a multidisciplinary approach that combines principles and practices from the fields of ***computer science***, ***statistics***, ***mathematics***, ***artificial intelligence***, ***machine learning***, and ***computer engineering*** to analyze large amounts of data.

Data Science helps people answer questions such as

- what happened, why it happened, what will happen, and what actions can be taken based on the results.



Why data science matters

World is generating more data than ever before

- The total amount of data stored by industry doubles every 1.2 years.

Raw data by itself is not useful

- Organizations need people who can **interpret** data for better decisions, better products, and better services.

Data science = language in new era

- Data science is the language of data, it helps us turn information into insight, decisions, and real-world impact.

Data science at URI

- provides students with the skills and understanding needed to produce valuable insights from large datasets

Every field uses data

Healthcare, business, public policy, sports, engineering, and science all rely on data to guide decisions.

Data must be interpreted

Information alone is not enough. Students need the skills to clean, analyze, model, and communicate what the data mean.

Why data science matters

It supports innovation

- Many modern technologies like AI, ML depend on data science (ex., AI grows by learning from data).

It creates strong career opportunities

- Data-related skills are in high demand because employers want people who can analyze data and communicate results clearly.

It combines multiple valuable skills

- Students learn statistics, computing, machine learning, AI, and domain knowledge, which makes them versatile.

Data science at URI

- Drawing on skills from computer science, mathematics, and statistics, **students learn** to use data to make decisions, discoveries, and predictions aimed at solving real-world problems.

Technology depends on it

AI, forecasting, recommendation systems, risk assessment, and many digital tools are built on data science ideas.

Strong versatility

These skills transfer across industries, which makes the field attractive for students who want flexibility.

Programs

We provide a variety of programs to meet students' needs

1. If your interest is in pushing the boundaries of what is possible in the data science field..

- The Bachelor of Science (B.S.) Degree (CS-STA track or MTH-AMS track)

2. If your interest is in preparing for a data-enabled career in business, healthcare, transportation, sports, finance, government, e-commerce, gaming, social media, insurance, or other domain..

- The Bachelor of Arts (B.A.) Degree
- Minor in Data Science

Programs

B.S. in Data Science

- Best for students who want a more technical or mathematically **intensive route**.
- **CS-STA track** emphasizes machine learning, statistics, and programming (Technical route).
(e.g., Programming, Statistical Computing, Artificial Intelligence, Machine Learning, Data Science Internship)
- **MTH-AMS track** emphasizes mathematical foundations and algorithmic thinking (Mathematically deeper route).
(e.g., Calculus for Functions, Predictive Analytics, Matrix Analysis, Optimization, Machine Learning, Probability theory, Mathematical Statistics)

B.S. - CS/STA track

Focuses on statistical and machine learning methods plus programming skills for a data science career.
Includes a required capstone experience (internship).

B.S. - MTH/AMS track

Focuses on the mathematical foundations behind advanced data analysis, algorithms, and machine learning methods.

Programs

B.A. in Data Science / Minor in Data Science

- Best for students who want data science **plus another primary field**.
- Examples include biology, business, communication studies, political science, psychology, and finance.
- Well suited for **double majors** and applied, domain-centered interests.
- Minor in D.S. requires minimum of 19 credits from the D.S. program (B.S. in D.S.: 56 credits, B.A. in D.S.: 29 credits).

B.A. in Data Science

Designed for students who want to double major or apply data science in another domain such as biology, business, communication, political science, psychology, or finance.

Minor in Data Science

A strong complement to nearly any major for students who want practical data fluency without changing majors.

Programs

Accelerated 5-year Bachelor and Master's degree (ABM)

- In just 5 years, students can earn a B.S. in Data Science and an M.S. in Mathematics, Computer Science, or Statistics.
- Students in the Data Science B.S. program with a certain academic achievement are eligible to apply.
- Students begin taking graduate courses in the spring semester of their third year.
- allow for 11 credits each to be double counted towards both the bachelor's degree and the master's degree.

ABM program

Designed for students who want a deeper understanding of the field and a master's degree to support their future careers.

Hands-on and Career-ready

Capstone experience

- BA/BS students are required (for Minor encouraged) to take **one class (research project or internship)** that is focused in applying data science principles/skills to a data intensive domain area.
- The capstone experience gives students the **opportunity to apply classroom knowledge to a real-world problem or project.**

Research with professors

- Students can conduct research with professors in Computer Science, Statistics, and Mathematics.

Internships

- Fidelity Investments, Johnson and Johnson, Merck, Blue Cross Blue Shield, Air National Guard, South County Health, General Dynamics Information Technology, The Hartford, ...

Career

Strong Career Prospects in a Data-Driven World

1. Grow 34% from 2024 to 2034

- According to the U.S. Bureau of Labor Statistics, employment of **data scientists** is projected to grow **34% from 2024 to 2034**, which is much faster than the average for all occupations. BLS also projects about **23,400 openings per year** on average over that decade.

2. 2024 median annual wage of \$112,590 for data scientists

- BLS reports a **2024 median annual wage of \$112,590** for data scientists. It also lists data scientists among the occupations with the **most new jobs**, with about **82,500 projected new jobs** from 2024 to 2034.

34%

Projected growth

Employment of data scientists is projected to grow from 2024 to 2034 — much faster than the average for all occupations.

23,400

Openings per year

About 23,400 openings for data scientists are projected each year, on average, over the decade.

\$112,590

Median annual pay

The median annual wage for data scientists was \$112,590 in May 2024, reflecting strong labor-market demand.

Career

Strong Career Prospects in a Data-Driven World

Broad industry demand

The outlook reflects needs across healthcare, finance, business, technology, science, public policy, and many other sectors.

Transferable skill set

Students build statistics, coding, data management, modeling, and communication skills that travel well across roles and industries.

Room to grow

A bachelor's degree can open doors early, and graduate study can deepen expertise for students who want more advanced technical or research paths.

Good fit for a changing economy

As AI and analytics become more common, data literacy becomes more valuable. **This major prepares students to work with those changes, not fear them.**

Take away!

This is **not only about one job title**. It is about helping students become fluent in data, computing, and evidence-based decision-making — skills with long-term value.

AI and Data Science

It is making data literacy, statistical thinking, and computational skills even more valuable

1. AI does not replace the need for data science—it makes it more important.

- AI systems depend on data, and organizations still need people who can collect, clean, evaluate, interpret, and responsibly use that data. The U.S. Bureau of Labor Statistics still projects **34% growth in data scientist jobs from 2024 to 2034**, which is much faster than average.

2. The best careers will belong to students who can work with AI, not compete against it.

- AI automates jobs. But, URI students are **not being trained for narrow repetitive tasks**; they are being trained to **use data and AI responsibly to solve problems**. BLS notes that while AI may reduce demand in some occupations, it can also **support demand for computer and data-related occupations** because organizations need people to build, maintain, and use more complex AI and data systems.

AI and Data Science

The goal is not to compete with AI, but to work effectively with it

3. Human judgment still matters.

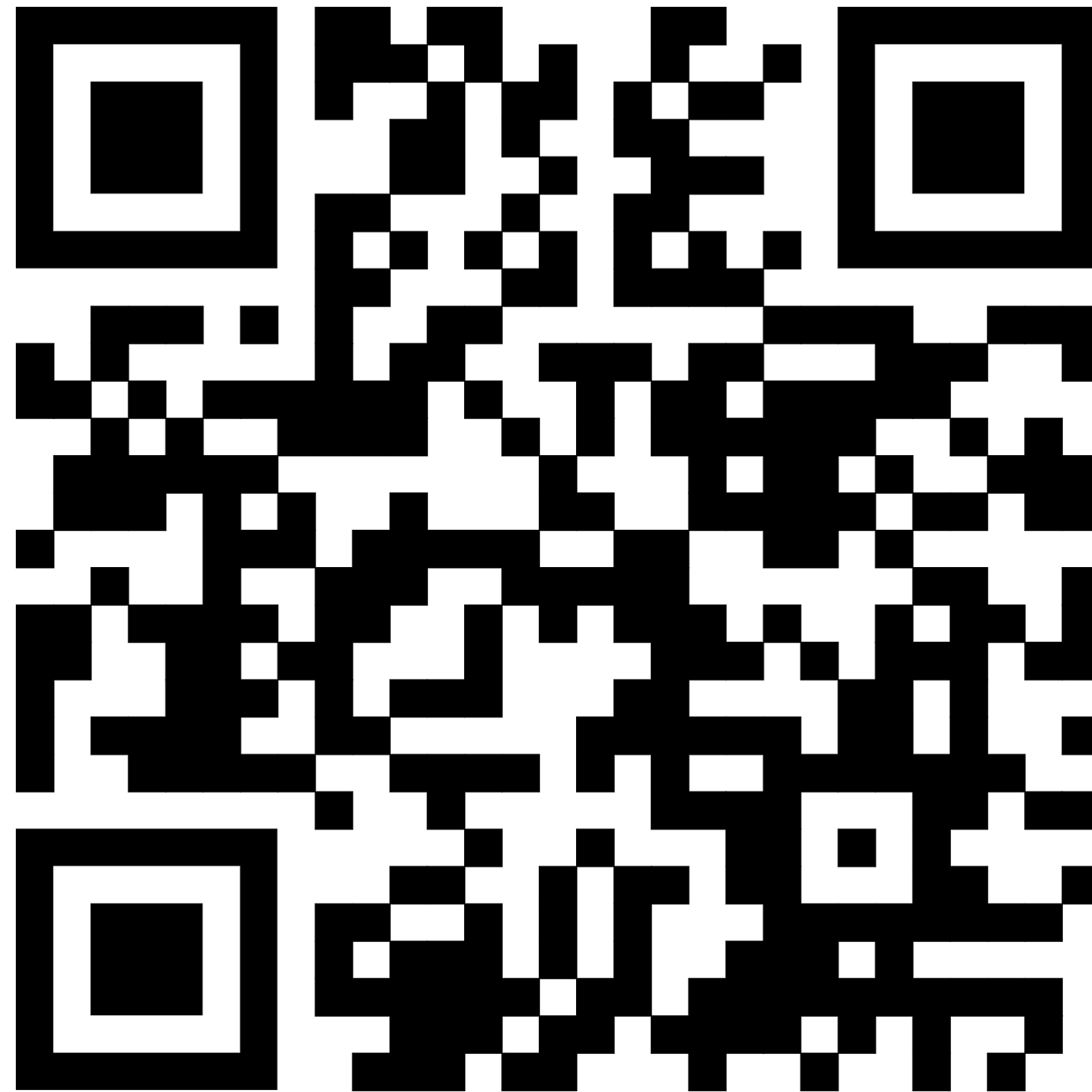
- Even in organizations using AI successfully, human validation, strategy, and domain understanding remain important. McKinsey's 2025 survey highlights that high-performing organizations are more likely to have processes for **human validation of model outputs**.

4. Data science teaches students how to evaluate and interpret AI

- We will continue to expand our AI-related course offerings, including Natural Language Processing, Generative AI, Neural Networks, Deep Learning, Reinforcement Learning, and Ethics in Artificial Intelligence.

Data science prepares students not just for today's tools, but for the next generation of AI-driven careers.

Thank you and welcome to URI!



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